

Course Syllabus

Course Information

Course Title: Statistics for Information Science

Course Number: INST314

Term: *Spring 2024*

Credits:3

Course Dates: January 24th - May 9th

Course Times: Lecture Sections 1-3 Time: Tu/Th 2:00 pm - 2:50 pm, Lecture Section 4 Time: Tu/Th 4:00 pm - 5:15 pm, Discussion 1: Time: Friday 1:00 pm - 1:50 pm, Discussion 2: Time: Friday 2:00 pm - 2:50 pm, Discussion 3: Time: Friday 3:00 pm - 3:50 pm

Professor: Samantha Kemper

Pronouns: She/Her

Office Hours: See home page

Email: skemper@umd.edu

Course Description

Basic concepts in statistics including measure construction, data exploration, hypothesis development, hypothesis testing, pattern identification, and statistical analysis. The course also provides an overview of commonly used data manipulation and analytic tools. Through homework assignments, projects, and in-class activities, you will practice working with these techniques and tools to create information resources that can be used in individual and organizational decision-making and problem-solving.

Prerequisite

Minimum grade of C- in [INST126](#)

[Links to an external site.](#)

or [GEOG276](#)

[Links to an external site.](#)

; and minimum grade of C- in [STAT100](#)

[Links to an external site.](#)

, and [MATH115](#)

[Links to an external site.](#)

(or higher).

Course Objectives

After successfully completing this course you will be able to:

1. Select and evaluate various types of data to use in decision making;
2. Use prescriptive and descriptive analyses to reach defensible, data-driven conclusions;
3. Select and apply appropriate statistical methods;
4. Use R for basic data manipulation and analysis;
5. Critically evaluate data analyses and develop strategies for making better decisions.

Required Resources

- Course Website: umd.edu
- [Links to an external site.](#)
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- Book: Links will be provided within modules to relevant sections in free, online resources
- A computer. Please contact me if you do not have access to a computer before class begins.
- R programming software (free, open source) <https://www.r-project.org/>
- [Links to an external site.](#)
- You may wish to install packages that will be used in-class. These will be linked in each module and will be available no later than Monday of each week. You can begin with the tidyverse using the command `install.packages("tidyverse")`. This may take 10-15 minutes to install. Wait until the command prompt returns before performing other actions in R.
- I strongly encourage you to use RStudio, which I will predominantly use during the class. It is an integrated development environment for R. (free and open source). You must download and install R prior to installing Rstudio. <https://www.rstudio.com/products/rstudio/download/#download>
- [Links to an external site.](#)
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- An alternative to installing R on your computer is to use RStudio Cloud. <https://rstudio.cloud/>
- [Links to an external site.](#)
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Course Structure

This will be a synchronous in-person course. You will need a computer with internet access that can log onto the ELMs course page. If you do not have access to a computer, please contact me before the beginning of class.

Each module will follow a similar set-up with assigned readings, lectures, or other videos to watch and engage with the material. Each module will have one or more homeworks accounting for 30% of the final grade in which students will practice the

skills covered in class. Weekly quizzes accounting for 10% of the final grade will cover key terms and concepts from the lecture and reading. These are intended as comprehension checks and are open note. The lowest quiz grade will be dropped. Class discussions and short group projects will constitute 15% of the final grade. Finally, three cumulative midterm projects worth 15% of the final grade each for a total of 45% of the final grade will take the place of exams.

If for any reason students need an extension on any of the assignments, please email me at skemper@umd.edu **before** the assignment is due, if possible. I realize that things can happen that interfere with a student's ability to learn and do their best work. Unless arrangements have been made with the instructor, all late work will accrue a 10% penalty per day and will no longer be accepted after 3 days. Late work will not be accepted after the assignment has been graded.

Tips for Success

1. **Participate.** Discussions and group work are a critical part of the course. You can learn a great deal from discussing ideas and perspectives with your peers and professor. Participation can also help you articulate your thoughts and develop critical thinking skills.
2. **Manage your time.** Make time for your online learning and participation in discussions each week. Give yourself plenty of time to complete assignments including extra time to handle any technology related problems.
3. **Login regularly.** Log in to ELMS-Canvas several times a week to view announcements, discussion posts and replies to your posts.
4. **Do not fall behind.** This class moves at a quick pace and each week builds on the previous. It will be hard to keep up with the course content if you fall behind in the pre-work or post-work.
5. **Use ELMS-Canvas notification settings.** Canvas ELMS-Canvas can ensure you receive timely notifications in your email or via text. Be sure to enable announcements to be sent instantly or daily.
6. **Ask for help if needed.** If you need help with ELMS-Canvas or other technology, IT Support. If Campus Policies

It is our shared responsibility to know and abide by the University of Maryland's policies that relate to all courses, which include topics like:

- Academic integrity

- Student and instructor conduct
- Accessibility and accommodations
- Attendance and excused absences
- Grades and appeals
- Copyright and intellectual property

Please visit www.ugst.umd.edu/courserelatedpolicies.html

[Links to an external site.](#)

for the Office of Undergraduate Studies' full list of campus-wide policies and follow up with me if you have questions.

Names/Pronouns and Self-Identifications

The University of Maryland recognizes the importance of a diverse student body, and we are committed to fostering inclusive and equitable classroom environments. I invite you, if you wish, to tell us how you want to be referred to both in terms of your name and your pronouns (he/him, she/her, they/them, etc.). The pronouns someone indicates are not necessarily indicative of their gender identity. Visit trans.umd.edu

[Links to an external site.](#)

to learn more.

Additionally, how you identify in terms of your gender, race, class, sexuality, religion, and dis/ability, among all aspects of your identity, is your choice whether to disclose (e.g., should it come up in classroom conversation about our experiences and perspectives) and should be self-identified, not presumed or imposed. I will do my best to address and refer to all students accordingly, and I ask you to do the same for all of your fellow Terps.

Communication with Instructor:

Email: If you need to reach out and communicate with me, please email me at skemper@umd.edu. Please DO NOT email me with questions that are easily found in the syllabus or on ELMS (i.e. When is this assignment due? How much is it worth? etc.) but please DO reach out about personal, academic, and intellectual concerns/questions.

While I will do my best to respond to emails within 24 hours, you will more likely receive email responses from me within 4 hours between 8 am and 10 pm EST on weekdays and within 24 hours on weekends.

ELMS: I will send IMPORTANT announcements via ELMS messaging. You must make sure that your email & announcement notifications (including changes in assignments and/or due dates) are enabled in ELMS so you do not miss any messages. You are responsible for checking your email and Canvas/ELMS inbox with regular frequency.

Communication with Peers:

With a diversity of perspectives and experience, we may find ourselves in disagreement and/or debate with one another. As such, it is important that we agree to conduct ourselves in a professional manner and that we work together to foster and preserve a virtual classroom environment in which we can respectfully discuss and deliberate controversial questions.

I encourage you to confidently exercise your right to free speech—bearing in mind, of course, that you will be expected to craft and defend arguments that support your position. Keep in mind, that free speech has its limit and this course is NOT the space for hate speech, harassment, and derogatory language. I will make every reasonable attempt to create an atmosphere in which each student feels comfortable voicing their argument without fear of being personally attacked, mocked, demeaned, or devalued.

Any behavior (including harassment, sexual harassment, and racially and/or culturally derogatory language) that threatens this atmosphere will not be tolerated. Please alert me immediately if you feel threatened, dismissed, or silenced at any point during our semester together and/or if your engagement in discussion has been in some way hindered by the learning environment.

Major Assignments

Homework Assignments

- Practice performing and interpreting analysis methods discussed in the synchronous lecture
- There will be roughly 1 homework per week

Quizzes

- Quick assessments of students' comprehension of major concepts and knowledge of key terms
- Administered within each module in ELMS. These are timed.

Discussions and Group Projects

- During live session and as part of the Discussion sections. Attendance at the class the discussion takes place is required for credit.
- Roughly 1 per module
- Designed to help students explore the broader implications of statistical use and misuse

Midterm Projects

- 3 cumulative projects
- Project 1 will be worth 10% of your final grade, Project 2 15% and Project 3 20% for a total of 45%

- Ask students to synthesize knowledge and skills learned thus far to choose the appropriate statistical tools to answer the questions, perform the analysis and interpret the results
- 1+ week to complete each project (later projects will be given more time)

Grading Structure

Assignment	Percentage %
Homework	30%
Quizzes	10%
Discussions/Group Projects	15%
3 Cumulative Midterm Projects	45%
Total	100%

Academic Integrity

The University's [Code of Academic Integrity](#)

[Links to an external site.](#)

is designed to ensure that the principles of academic honesty and integrity are upheld. In accordance with this code. Please ensure that you fully understand this code and its implications because all acts of academic dishonesty will be dealt with in accordance with the provisions of this code. All students are expected to adhere to this Code. It is your responsibility to read it and know what it says, so you can start your professional life on the right path. **As future professionals, your commitment to high ethical standards and honesty begins with your time at UMD.**

It is important to note that course assistance websites, such as CourseHero, are not permitted sources, unless the instructor explicitly gives permission for you to use one of these sites. Material taken or copied from these sites can be deemed unauthorized material and a violation of academic integrity. These sites offer information that might not be accurate and that shortcut the learning process, particularly the critical thinking steps necessary for college-level assignments.

Additionally, it is understandable that students may use a variety of online or virtual forums for course-wide discussion (e.g., GroupME or WeChat). Collaboration in this way regarding concepts discussed in this course is permissible. However, collaboration on graded assignments is strictly prohibited unless otherwise stated.

Please visit the [Office of Undergraduate Studies' full list of campus-wide policies](#)

[Links to an external site.](#)

and follow up with me if you have questions.

To help you avoid unintentional violations, ***the following table*** lists levels of collaboration that are acceptable for each type of assignment. If you ever feel pressured to comply with someone else's academic integrity violation, please reach out to me straight away. Also, ***if you are ever unclear*** about acceptable levels of collaboration, ***please ask!***

Grades

Your grade will reflect your mastery of the material. There is no effort or participation grade. These are skills that will be needed for all of you as working professionals, whether as someone who generates the analysis or as someone who consumes analyses. If life circumstances are making it difficult to perform at your best, please contact me. I will do my best to work with you.

Scores for all assignments will be posted on the course ELMS page. If you would like to review any of your grades or have questions about how something was scored, please email me to schedule a time for us to meet and discuss. I will include comments on what was missed for the vast majority of grades on the assignments themselves. Please review these when your grades are posted.

Final letter grades are assigned based on the percentage of total points earned using the grading structure explained above. +/- grading will be used. Final grade cutoffs will be assigned based on the table below. To be fair to everyone I must establish clear standards and apply them consistently, so please understand that being close to a cutoff is not the same as making the cut ($89.99 \neq 90.00$). It would be unethical to make exceptions for some and not others.

Final Grade Cutoffs

+	97.00%	+	87.00%	+	77.00%	+	67.00%	+	
A	94.00%	B	84.00%	C	74.00%	D	64.00%	F	<60.0%
-	90.00%	-	80.00%	-	70.00%	-	60.00%	-	

Resources &

Accommodations

Accessibility and Disability

Services

The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational, working, and living environment for people of all abilities. The University of Maryland is also committed to the principle that no qualified individual with a disability shall, on the basis of disability, be excluded from participation in or be denied the benefits of the services, programs, or activities of the University, or be subjected to discrimination. The [Accessibility & Disability Service \(ADS\)](#)

[Links to an external site.](#)

provides reasonable accommodations to qualified individuals to provide equal access to services, programs and activities. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact me as soon as possible so that I have sufficient time to make arrangements.

For assistance in obtaining an accommodation, contact Accessibility and Disability Service at 301-314-7682, or email them at adsfrontdesk@umd.edu. Information about [sharing your accommodations with instructors](#)

[Links to an external site.](#)

, [note taking assistance](#)

[Links to an external site.](#)

and more is available from the [Counseling Center](#)

[Links to an external site.](#)

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Student Resources and Services

Taking personal responsibility for your own learning means acknowledging when your performance does not match your goals and doing something about it. I hope you will come talk to me so that I can help you find the right approach to success in this course, and I encourage you to visit [UMD's Student Academic Support Services website](#)

[Links to an external site.](#)

to learn more about the wide range of campus resources available to you.

In particular, everyone can use some help sharpen their communication skills (and improving their grade) by visiting [UMD's Writing Center](#)

[Links to an external site.](#)

and schedule an appointment with the campus Writing Center.

You should also know there are a wide range of resources to support you with whatever you might need ([UMD's Student Resources and Services website](#)

[Links to an external site.](#)

may help). If you feel it would be helpful to have someone to talk to, visit [UMD's Counseling Center](#)

[Links to an external site.](#)

or [one of the many other mental health resources on campus](#)

[Links to an external site.](#)

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Basic Needs Security

If you have difficulty affording groceries or accessing sufficient food to eat every day, or lack a safe and stable place to live, please visit [UMD's Division of Student Affairs website](#)

[Links to an external site.](#)

for information about resources the campus offers you and let me know if I can help in any way.

Netiquette Policy

Netiquette is the social code of online classes. Students share a responsibility for the course's learning environment. Creating a cohesive online learning community requires learners to support and assist each other. To craft an open and interactive online learning environment, communication has to be conducted in a professional and courteous manner at all times, guided by common sense, collegiality and basic rules of etiquette.

Participation

- Given the interactive style of this class, attendance will be crucial to note-taking and thus your performance in this class. Attendance is particularly

important also because class discussion will be a critical component for your learning.

- Each student is expected to make substantive contributions to the learning experience, and attendance is expected for every session.
- Students with a legitimate reason to miss a live session should communicate in advance with the instructor, except in the case of an emergency.
- Students who miss a live session are responsible for learning what they miss from that session.
- Additionally, students must complete all readings and assignments in a timely manner in order to fully participate in class.

Course Evaluation

Please submit a course evaluation through CourseEvalUM in order to help faculty and administrators improve teaching and learning at Maryland. All information submitted to CourseEvalUM is confidential. Campus will notify you when CourseEvalUM is open for you to complete your evaluations for fall semester courses. Please go directly to the [Course Eval UM website](#)

[Links to an external site.](#)

to complete your evaluations. By completing all of your evaluations each semester, you will have the privilege of accessing through Testudo, the evaluation reports for the thousands of courses for which 70% or more students submitted their evaluations.

Copyright Notice

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